



Minterm and Maxterm

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Standard Forms

- Facilitate simplification
- Result in more desirable implementations
- Standard Forms rely on two type of terms
 - *Product Terms* – Terms that are ANDed together
 - XYZ
 - $(A+B)(C+D)(A+D)$
 - *Sum Terms* – Terms that are ORed together
 - $X+Y+Z$
 - $XYZ + VX$

Minterms

- Are AND terms with every variable presented in either un-complemented (true) or complement form $(x, \text{ or } x')$
- For Minterms :
 - “1” $\rightarrow$ means the variable is true (not complemented) (x)
 - “0” $\rightarrow$ means the variable is complemented (x')

Minterm for 3 variables

Row number	x_1	x_2	x_3	Minterm
0	0	0	0	$m_0 = \overline{x}_1 \overline{x}_2 \overline{x}_3$
1	0	0	1	$m_1 = \overline{x}_1 \overline{x}_2 x_3$
2	0	1	0	$m_2 = \overline{x}_1 x_2 \overline{x}_3$
3	0	1	1	$m_3 = \overline{x}_1 x_2 x_3$
4	1	0	0	$m_4 = x_1 \overline{x}_2 \overline{x}_3$
5	1	0	1	$m_5 = x_1 \overline{x}_2 x_3$
6	1	1	0	$m_6 = x_1 x_2 \overline{x}_3$
7	1	1	1	$m_7 = x_1 x_2 x_3$

Minterms for n variable functions

- For 2 variables have 4 minterms
 - $X'Y'$ $X'Y$ XY' XY
- For 3 variables have 8 minterms
 - $X'Y'Z'$ $X'Y'Z$... XYZ
- In general, if a function has n variables there are 2^n minterms

Maxterms

- Are OR terms with every variable in complemented (true) or un-complemented form (x , or x')
- For Maxterms:
 - “0” $\rightarrow$ means the variable is true (not complemented) (x)
 - “1” $\rightarrow$ means the variable is complemented (x')

Maxterm and Minterm table

Row number	x_1	x_2	x_3	Minterm	Maxterm
0	0	0	0	$m_0 = \bar{x}_1 \bar{x}_2 \bar{x}_3$	$M_0 = x_1 + x_2 + x_3$
1	0	0	1	$m_1 = \bar{x}_1 \bar{x}_2 x_3$	$M_1 = x_1 + x_2 + \bar{x}_3$
2	0	1	0	$m_2 = \bar{x}_1 x_2 \bar{x}_3$	$M_2 = x_1 + \bar{x}_2 + x_3$
3	0	1	1	$m_3 = \bar{x}_1 x_2 x_3$	$M_3 = x_1 + \bar{x}_2 + \bar{x}_3$
4	1	0	0	$m_4 = x_1 \bar{x}_2 \bar{x}_3$	$M_4 = \bar{x}_1 + x_2 + x_3$
5	1	0	1	$m_5 = x_1 \bar{x}_2 x_3$	$M_5 = \bar{x}_1 + x_2 + \bar{x}_3$
6	1	1	0	$m_6 = x_1 x_2 \bar{x}_3$	$M_6 = \bar{x}_1 + \bar{x}_2 + x_3$
7	1	1	1	$m_7 = x_1 x_2 x_3$	$M_7 = \bar{x}_1 + \bar{x}_2 + \bar{x}_3$

Specifying functions

Any boolean Function can be expressed in minterm or maxterm notation

- SOP (sum of product)
- POS (product of sum)

SOP (sum of product)

- Equations are written as an OR of AND terms

$$f(A,B,C) = ABC + A'BC'$$

Sum

Product terms

POS (product of sum)

- Equations are written as an AND of OR terms

$$f(A,B,C) = (A' + B) \cdot (B + C')$$

Product

Sum terms

examples

- express the **Boolean function** :

$$F(X, Y, Z) = \sum (0, 2, 5, 7)$$

Answer:

Function exist at minterms :

$$\rightarrow m_0 + m_2 + m_5 + m_7$$

“1” → means the variable is true (not complemented) (x)

“0” → means the variable is complemented (x')

$$F(X, Y, Z) = X'Y'Z' + X'YZ' + XY'Z + XYZ$$

examples

find the minterms of this Boolean function:

$$F(X,Y,Z) = X'Y'Z' + X'YZ' + XY'Z + XYZ$$

Answer:

“1” → means the variable is true (not complemented) (x)

“0” → means the variable is complemented (x')

$$F(X, Y, Z) = 000 + 010 + 101 + 111$$

Function exist at :

$$F(X, Y, Z) = \sum (0, 2, 5, 7)$$

examples

express the **Boolean function** :

$$F(X, Y, Z) = \prod (0, 2, 4, 5)$$

Answer:

Function exist at maxterm:

$$\rightarrow M_0 + M_2 + M_4 + M_5$$

“0” $\rightarrow$ means the variable is true (not complemented) (x)

“1” $\rightarrow$ means the variable is complemented (x')

$$F(X, Y, Z)$$

$$= (X + Y + Z). (X + Y' + Z). (X' + Y + Z). (X' + Y + Z')$$

Examples

Find the maxterms of this Boolean function:

$$F(X, Y, Z)$$

$$= (X + Y + Z). (X + Y' + Z). (X' + Y + Z). (X' + Y + Z')$$

Answer:

“0” → means the variable is true (not complemented) (x)

“1” → means the variable is complemented (x')

$$F(X, Y, Z) = 000 + 010 + 100 + 101$$

Function exist at :

$$F(X, Y, Z) = \prod(0, 2, 4, 5)$$

Conversion between SOP and POS

- The complement of sum of minterms functions (F) equals the sum of minterms missing from the original function (F').
- If we take the complement of F' using DeMorgan's theorem, we obtain the product of maxterms function
- $m'_j = M_j$

example

Find the maxterm of this Boolean function:

$$F(A, B, C) = \sum (1, 2, 4, 5, 6, 7)$$

Answer:

- Complement of F $\rightarrow$ the missing minterms

$$F'(A, B, C) = \sum (0, 2, 3) = m_0 + m_2 + m_3$$

By using DeMorgan's theorem:

$$(m_0 + m_2 + m_3)' = M_0 \cdot M_2 \cdot M_3$$

So, function found at maxterms:

$$F(A, B, C) = \prod (0, 2, 3)$$

Another example

Express the function in minterm notation

$$F(A, B, C) = AB + A'C$$

- First find the missing literal in each product term expand to where each term has all variables in it.
- AND term with 1 to expand.
For the 1st term c is missing AND with $(C+C')$
for the 2nd the B is missing AND with $(B+B')$

- Now have

$$F(A, B, C) = AB(C + C') + A'C(B + B')$$

$$F(A, B, C) = ABC + ABC' + A'BC + A'B'C$$

$$= m_7 + m_6 + m_3 + m_1$$

$$F(A, B, C) = \sum (1, 3, 6, 7)$$

Express the function in minterm notation

$$F(X, Y, Z) = Y' + X'Z'$$

Answer:

$$\begin{aligned} F(X, Y, Z) &= (X + X')Y' + X'Z'(Y + Y') \\ &= XY'(Z + Z') + X'Y'(Z + Z') + X'YZ' + X'Y'Z' \\ &\quad XY'Z + XY'Z' + X'Y'Z + X'Y'Z' + X'YZ' \\ &= m_5 + m_4 + m_1 + m_0 + m_2 \\ F(X, Y, Z) &= \sum (0, 1, 2, 4, 5) \end{aligned}$$

Find the Minterm Expansion

- $f(a, b, c, d) = a'(b' + d) + acd'$

Answer:

$$a'b' + a'd + acd'$$

$$= a' b' (c + c') (d + d') + a' d (b + b') (c + c') + acd' (b + b')$$

$$= a' b' c' d' + a' b' c' d + a' b' c d' + a' b' c d + a' b c' d + a' b c d + a b c d' + a b' c d'$$

$$= \sum (0, 1, 2, 3, 5, 7, 10, 14)$$

Find the Maxterm Expansion

$$f(a, b, c, d) = a'(b' + d) + acd'$$

Answer:

$$= (a' + cd')(a + b' + d)$$

Use $(x + y)(x' + z) = (xz + x'y)$

$$(a' + c)(a' + d')(a + b' + d)$$

Use $(x + y)(x + z) = (x + yz)$

$$= (a' + bb' + c + dd')(a' + bb' + cc' + d')(a + b' + cc' + d)$$

$$= (a' + bb' + c + d)(a' + bb' + c + d') (a' + bb' + c + d') (a' + bb' + c' + d') (a + b' + cc' + d)$$

$$= (a' + b + c + d)(a' + b' + c + d)(a' + b + c + d') (a' + b' + c + d') (a' + b + c' + d') (a' + b' + c' + d') (a + b' + c + d)(a + b' + c' + d)$$

$$f(a, b, c, d) = \prod (4, 6, 8, 9, 11, 12, 13, 15)$$

Or we can solve this problem with complement of DeMorgan's theorem.

1- first solve the problem using Minterm

Ex: $f(a, b, c, d) = a'(b' + d) + acd'$

ANS:

$$f(a, b, c, d) = \sum (0, 1, 2, 3, 5, 7, 10, 14)$$

2- find the complement of function (missing minterms)

$$f'(a, b, c, d) = \sum (4, 6, 8, 9, 11, 12, 13, 15, 16)$$

3- use DeMorgan to find the complement

$$f(a, b, c, d) = \prod (4, 6, 8, 9, 11, 12, 13, 15, 16)$$

Express the Boolean function as a product of Maxterms

- $F(x, y, z) = xy + x'z$

Answer:

$$= xy(z + z') + x'z(y + y')$$

$$= xyz + xyz' + x'yz + x'y'z$$

$$F(x, y, z) = m_7 + m_6 + m_3 + m_1$$

$$F(x, y, z) = \sum (1, 3, 6, 7)$$

In maxterm as follows:

$$F(x, y, z) = \prod (0, 2, 4, 5)$$

- Example:
- Find the minterm and maxterm of these Boolean functions :

$$F_1 = x'y'z + xy'z' + xyz$$

$$F_2 = x'yz + xy'z + xyz' + xyz$$

Answer:

$$F_1 = \sum (1,4,7)$$

$$F_2 = \sum (3,5,6,7)$$

	x	y	z	F_1	F_2
m_0	0	0	0	0	0
m_1	0	0	1	1	0
m_2	0	1	0	0	0
m_3	0	1	1	0	1
m_4	1	0	0	1	0
m_5	1	0	1	0	1
m_6	1	1	0	0	1
m_7	1	1	1	1	1

- $F_1 = m_1 + m_4 + m_7$

Or

$$F_1 = M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6$$

- $F_2 = m_3 + m_5 + m_6 + m_7$

Or

$$F_2 = M_0 \cdot M_1 \cdot M_2 \cdot M_4$$

Prove that :

- $F_1 = x'y'z + xy'z' + xyz = M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6$

Answer:

the complement of F_1 is :

$$F'_1 = x'y'z' + x'yz' + x'yz + xy'z + xyz'$$

$$\begin{aligned} F_1 &= (x + y + z)(x + y' + z)(x + y' + z')(x' + y \\ &\quad + z')(x' + y' + z) \end{aligned}$$

$$= M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6$$